

JKBOSE 2026

CLASS 11TH

STRAIGHT LINES

MATHS

LECTURE-01

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TOPICS TO BE COVERED

1

Basics of Coordinate Geometry

2

Related Problems

INTRODUCTION

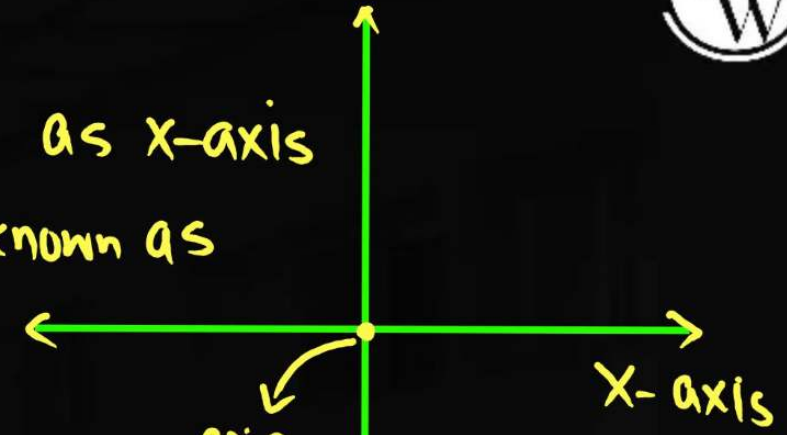
Co-ordinate Geometry:

- * It is a Branch of Mathematics in which we study Geometric problems through algebra.
- * It is a Marriage b/w Geometry and Algebra.
- * Rene Descartes is known as the Father of Co-ordinate Geometry.

* Co-ordinate axes:

- Set of perpendicular number lines are known as coordinate axes

- Horizontal line is known as x-axis & the vertical line is known as y-axis



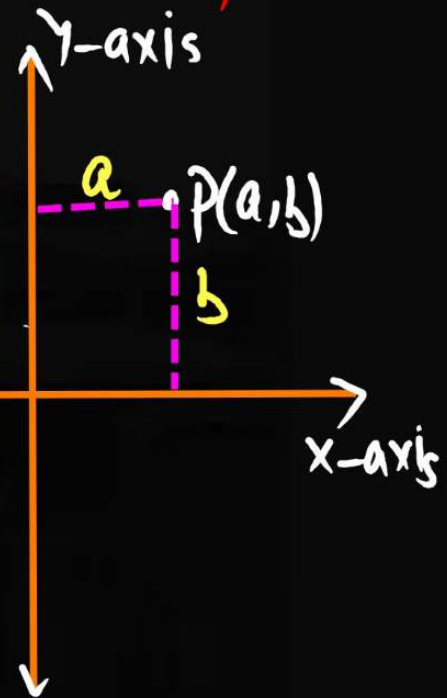
- Their point of intersection is known as origin

- Divides the plane into four parts (Quadrants)

* Co-ordinates of a Point:

- a is the distance of point P from y-axis, & is known as Abscissa or x-co-ordinate.

- b is the distance of point P from the x-axis, & is known as ordinate or y-co-ordinate.

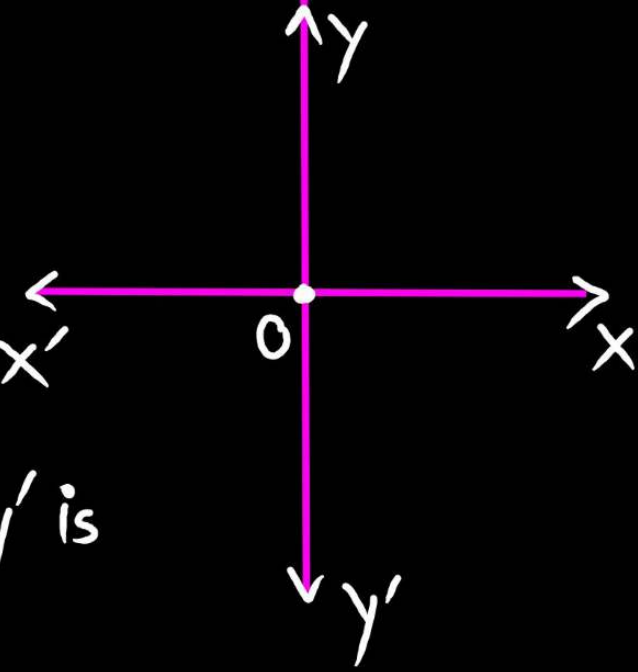


* Sign Conventions:

- Distance along OX & OY is taken as positive.
- Distance along OX' & OY' is taken as negative.

So, for any point $P(a, b)$, we have,

- $a > 0, b > 0$ for 1st Quadrant.
- $a < 0, b > 0$ for 2nd Quadrant.
- $a < 0, b < 0$ for 3rd Quadrant.
- $a > 0, b < 0$ for 4th Quadrant.



Question: The point $P(-2, 3)$ lies in,

- (a) 1st Quadrant ~~X~~ (c) 3rd Quadrant ~~X~~
 (b) 2nd Quadrant (d) 4th Quadrant ~~X~~

* Distance formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (\text{Distance formula})$$

* Section formula:

$$x = \frac{mx_2 + nx_1}{m+n}$$

$$y = \frac{my_2 + ny_1}{m+n} \Rightarrow C(x, y) = C\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$$



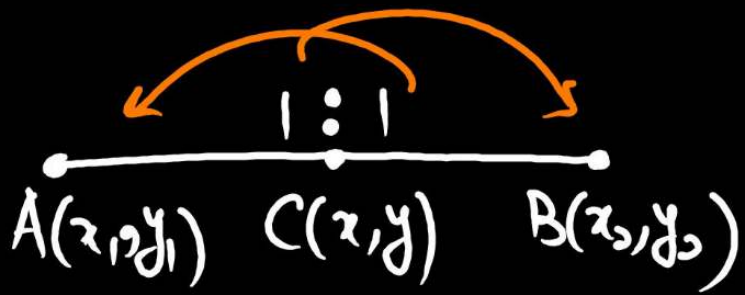
Mid-point formula:

By section formula;

$$x = \frac{x_2 + x_1}{2} = \frac{x_1 + x_2}{2}$$

$$y = \frac{y_2 + y_1}{2} = \frac{y_1 + y_2}{2}$$

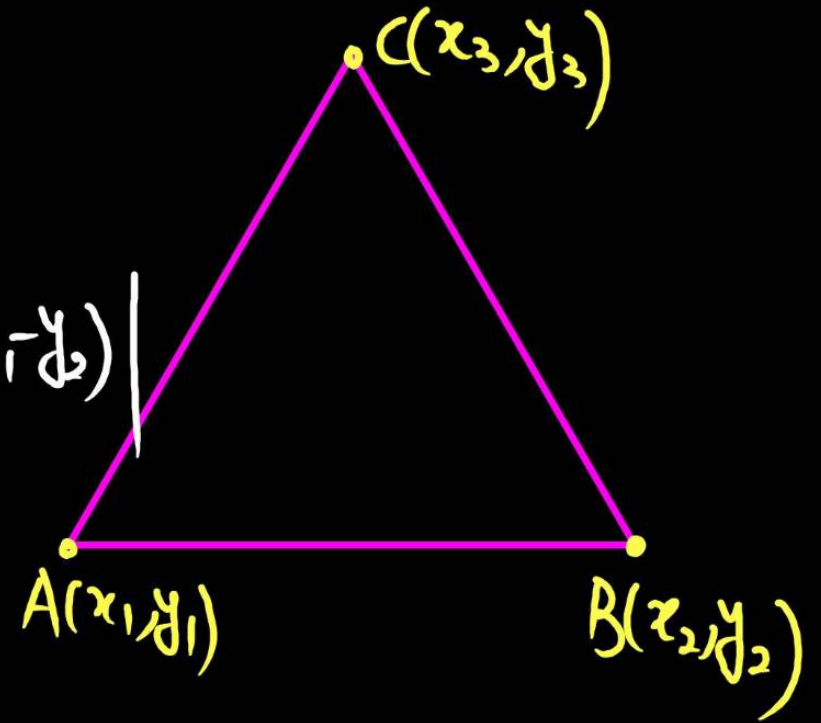
$$\Rightarrow C(x, y) = C\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$



* Area of Triangle:

$$Ar(\Delta ABC) = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

± If $Ar(\Delta ABC) = 0$, then we say that the points are collinear.



QUESTION

The distance between the Points **P(2, 3)** and **Q(-1, 2)** is :

1 $\sqrt{10}$ ✓

2 $\sqrt{13}$ ✗

3 $\sqrt{2}$ ✗

4 $\sqrt{8}$ ✗



$$PQ = \sqrt{(2+1)^2 + (3-2)^2}$$

$$\Rightarrow PQ = \sqrt{9+1} = \sqrt{10}$$

EXAMPLE

If the distance between the Points $P(-3, 5)$ and $Q(-x, -2)$ is $\sqrt{58}$ then find the value(s) of x .

(Sol) Given that $PQ = \sqrt{58}$

$$\Rightarrow \sqrt{(-3+x)^2 + (5+2)^2} = \sqrt{58}$$

$$\Rightarrow (x-3)^2 + 49 = 58$$

$$\Rightarrow (x-3)^2 = 58 - 49$$

$$\Rightarrow (x-3)^2 = 9$$

$$\Rightarrow x-3 = \pm 3$$

$$\Rightarrow x-3 = 3, -3$$

$$\Rightarrow x = 3+3, -3+3 \quad P(-3, 5)$$

$$\Rightarrow x = 6, 0$$

Hence the result.

$\sqrt{58}$

$Q(-x, -2)$

EXAMPLE

A line segment is of the length **15 units** and one end is at the point $(3, 2)$ if the **abscissa** of the other end is **15**, then find the possible ordinates

(Sol) Given that $AB = 15$

$$\Rightarrow \sqrt{(15-3)^2 + (y-2)^2} = (15)^2$$

$$\Rightarrow 144 + (y-2)^2 = 225$$

$$\Rightarrow (y-2)^2 = 225 - 144$$

$$\Rightarrow (y-2)^2 = 81$$

$$\Rightarrow y-2 = \pm 9$$

$$\rightarrow y-2 = 9, -9$$

$$\Rightarrow y = 9+2, -9+2$$

$$\Rightarrow y = 11, -7$$

Hence the result.



QUESTION

$$c < 3$$
$$c^2 < 9$$

$$c^2 - 9 < 0$$



The number of points on **x-axis** which are at a distance **c ($c < 3$)** from the point **$(2, 3)$** is :

1 $2 \times$

2 $1 \times$

3 infinite $\times$

4 No point $\checkmark$

(Sol) By Distance formula:

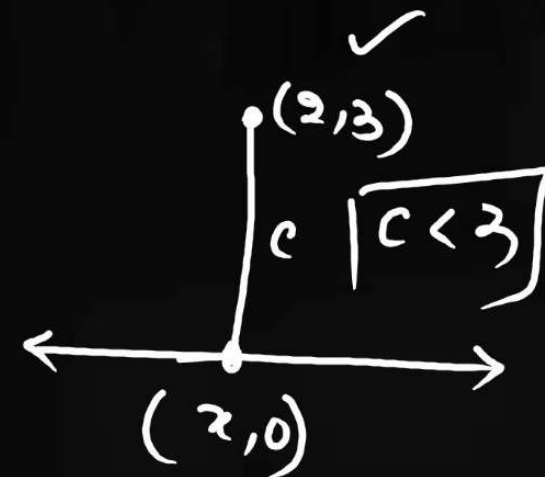
$$\sqrt{(x-2)^2 + (0-3)^2} = c \quad (c < 3)$$

$$\Rightarrow (x-2)^2 + 9 = c^2$$

$$\Rightarrow (x-2)^2 = c^2 - 9 < 0$$

$$\Rightarrow (x-2)^2 < 0, \text{ which is not possible.}$$

Hence there is no such point



QUESTION

$$\sqrt{x^2} = |x| = x, x \geq 0.$$



The distance between the point $P(a \cos \alpha, a \sin \alpha)$ and $Q(a \cos \beta, a \sin \beta)$, where $a > 0$ and $\alpha > \beta$, is :

1 $\times$ $4 a \sin \frac{\alpha - \beta}{2}$

2 $\times$ $4 a \sin \frac{\alpha + \beta}{2}$

3 $\checkmark$ $2 a \sin \frac{\alpha - \beta}{2}$

4 $\times$ $2 a \cos \frac{\alpha - \beta}{2}$

(Sol) Distance formula:

$$PQ^2 = (a \cos \alpha - a \cos \beta)^2 + (a \sin \alpha - a \sin \beta)^2$$

$$\Rightarrow PQ^2 = a^2 (\cos \alpha - \cos \beta)^2 + a^2 (\sin \alpha - \sin \beta)^2$$

$$\Rightarrow PQ^2 = a^2 \left(-2 \sin \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right) \right)^2 + a^2 \left(2 \cos \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right) \right)^2$$

$$\Rightarrow PQ^2 = 4a^2 \sin^2 \left(\frac{\alpha + \beta}{2} \right) \sin^2 \left(\frac{\alpha - \beta}{2} \right) + 4a^2 \cos^2 \left(\frac{\alpha + \beta}{2} \right) \sin^2 \left(\frac{\alpha - \beta}{2} \right)$$

$$\Rightarrow PQ^2 = 4a^2 \sin^2 \left(\frac{\alpha - \beta}{2} \right) \underbrace{\left[\sin^2 \left(\frac{\alpha + \beta}{2} \right) + \cos^2 \left(\frac{\alpha + \beta}{2} \right) \right]}_1$$

$P(a \cos \alpha, a \sin \alpha)$ $Q(a \cos \beta, a \sin \beta)$

$$\Rightarrow PQ^2 = \left(2a \sin \left(\frac{\alpha - \beta}{2} \right) \right)^2$$

$$\Rightarrow PQ = \sqrt{\left(2a \sin \left(\frac{\alpha - \beta}{2} \right) \right)^2}$$

$$\Rightarrow PQ = 2a \sin \left(\frac{\alpha - \beta}{2} \right)$$

EXAMPLE

Determine the ratio in which $y - x + 2 = 0$ divides the line joining $(3, -1)$ and $(8, 9)$.

(Sol) Suppose that the line $y - x + 2 = 0$ — ①

divides AB in the ratio $k:1$.

Then by section formula, we have,

$\left(\frac{8k+3}{k+1}, \frac{9k-1}{k+1}\right)$. Since the point C lies on ①

$$\text{So, } \left(\frac{9k-1}{k+1}\right) - \left(\frac{8k+3}{k+1}\right) + 2 = 0$$

$$\Rightarrow \frac{9k-1-8k-3}{k+1} + 2 = 0$$

$$\frac{k-4}{k+1} + 2 = 0$$

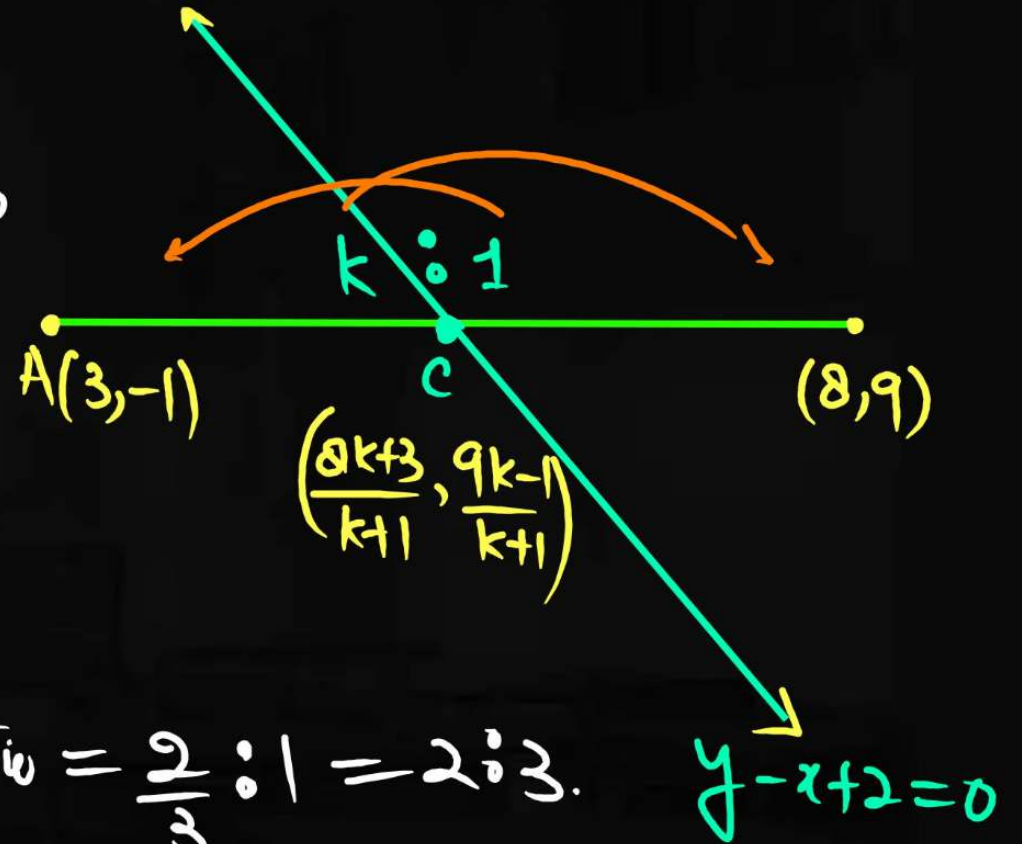
$$\Rightarrow k-4+2k+2=0$$

$$\Rightarrow 3k-2=0$$

$$\Rightarrow k = \frac{2}{3}$$

$$\text{So Required ratio} = \frac{2}{3} : 1 = 2 : 3.$$

Hence the result



QUESTION

If the vertices of a triangle are $(1, 2)$, $(4, -6)$ and $(3, 5)$, then its area is :

☒ $\frac{25}{2}$ sq. units

☐ 12 sq. units

☐ 5 sq. units

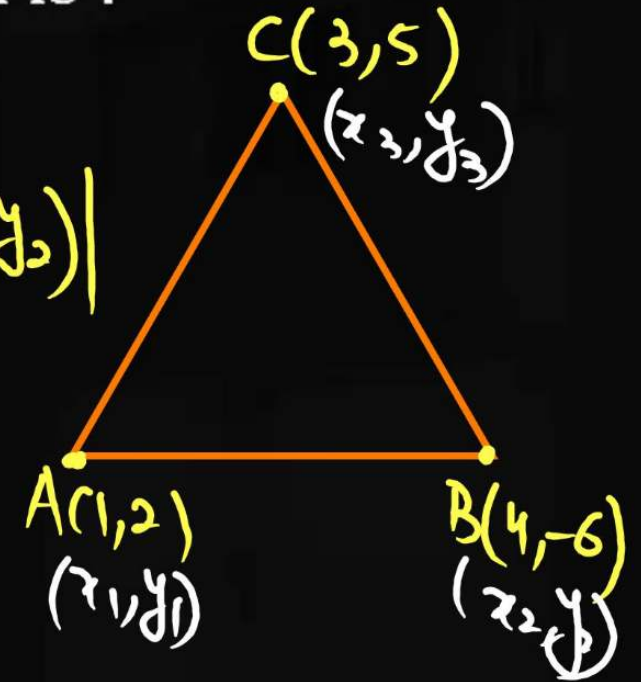
☐ 25 sq. units

(Sol) Here $Ar(\triangle ABC) = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$

$\Rightarrow Ar(\triangle ABC) = \frac{1}{2} |1(-6 - 5) + 4(5 - 2) + 3(2 + 6)|$

$\Rightarrow Ar(\triangle ABC) = \frac{1}{2} |-11 + 12 + 24|$

$\Rightarrow Ar(\triangle ABC) = \frac{1}{2} |25| = \frac{25}{2}$ Sq. units



QUESTION

The points A divides the join of the points $(-5, 1)$ and $(3, 5)$ in the ratio of $k : 1$ and coordinates of B and C are $(1, 5)$ and $(7, -2)$ respectively. If the area of ΔABC be 2 units, then k equals-

Homework
#

1 7, 9

2 6, 7

3 $7, \frac{31}{9}$

4 $9, \frac{31}{9}$

Thank
You